

SAFIRA RAHMALIA PUTRI

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Data Science graduate with a background in statistics, machine learning, and data analysis. Comfortable using Python, R, SQL, and Excel to clean data, explore it, build visualizations, and develop predictive models. I like working with data from start to finish, from cleaning and exploring it to building models and presenting the results in a way that is easy to follow. I pay attention to the details, because small mistakes in data can change the whole conclusion. I work well both independently and in a team, and I enjoy learning new tools whenever a problem needs them.

EDUCATION

Universitas Pembangunan Nasional “Veteran” Jawa Timur - Surabaya, Indonesia

Aug 2022 – Sep 2026

Undergraduate, Data Science, 3.79/4.00

INTERN EXPERIENCE

PT Pos Indonesia (Persero), Krembangan - Surabaya, Indonesia

Aug 2025

Information Technology (IT)

- Processed and organized assistance program data for reporting needs.
- Processed and compiled stamp duty (materai) purchase data.

BADAN PUSAT STATISTIK PROVINSI Jawa Timur - Surabaya, Indonesia

Sep 2024 – Dec 2024

Bidang Integrasi Pengolahan dan Diseminasi Statistik (IPDS)

- Processed and analyzed data for statistical publications, including educational data for schools across East Java.
- Collaborated with the IPDS team to streamline data management workflows and improve dissemination processes.

PUBLICATIONS AND PRESENTATIONS

“Automated Stroke Lesion Severity Estimation from Brain MRI Using Attention-Enhanced DenseNet121.”

Presented, Sep 2026

- Event: Presenter, 11th International Seminar of Research Month (ISRM) 2026, UPN “Veteran” Jawa Timur, Surabaya (September 22-23, 2026).
- Objective: Developed a pipeline that classifies stroke on brain MRI, segments the lesion, and estimates severity.
- Results: The best configuration achieved 84.65% accuracy, 0.8598 F1-score, and 0.9195 AUC-ROC, while NVAutoNet segmentation achieved a mean Dice of 0.8347.

“Effects of Modality Representation and Slice Selection Strategy on DenseNet121-CBAM Based Ischemic Stroke Lesion Classification and Lesion Volume Quantification.”

Under Review, 2026

- Journal: Journal of Advances in Information Technology. (SCOPUS)
- Objective: Examined how MRI modality representation and slice selection strategy affect DenseNet121-CBAM classification, and quantified lesion volume.
- Results: The best configuration achieved 84.65% accuracy, 0.8598 F1-score, and 0.9195 AUC-ROC, while NVAutoNet segmentation achieved a mean Dice of 0.8347.

“Classification of Stroke Lesions in MRI Images Using DenseNet121-CBAM.”

Accepted, 2026

- Journal: International Journal of Computer, Network Security and Information System (IJCONSIST). Letter of acceptance received, forthcoming (SINTA 4).
- Objective: Classified stroke lesions on brain MRI slices using DenseNet121 with CBAM attention.
- Results: The best configuration achieved an F1-score of 0.8598 and AUC of 0.9195.

“Classification of Stroke Lesions in MRI Images Using DenseNet121.”

2025

- Proceedings: Innovation in Numerical Technology, Engineering, Research, and Applied Learning (INT), Vol. 1 No. 1, 2025. Published.
- Objective: Developed a brain MRI stroke classification model using DenseNet121 as the feature extractor.
- Results: Validation accuracy improved from 87.62% to 97.25% through fine-tuning, with average precision, recall, and F1-score of 0.97.

“Implementation of a BiLSTM-Attention Model for IHSG Prediction Based on Historical OHLCV Data.”

2025

- Journal: Jurnal Ilmiah Media Sisfo, 19(2), 98-108 (SINTA 4).
- Objective: Developed a Bidirectional LSTM with an attention mechanism to predict the Indonesian Composite Stock Price Index (IHSG) from historical OHLCV data.
- Results: Reached R^2 0.9485, RMSE 59.47, MAE 45.12, and MAPE 0.63%. Attention weights showed the model relies most on the last two days of the input window.

“Analysis of the Use of the Words ‘Gadget’ and ‘Gawai’ in the Context of Era 4.0 to 5.0 Using Natural Language Processing (NLP) Methods and Naive Bayes Classification.”

Journal: Jurnal Pendidikan Tambusai, 8(2), 27912-27921. Published (SINTA 6)

Objective: Analyzed the usage of “gadget” and “gawai” across Technology 4.0 to 5.0 using NLP and Naive Bayes.

Results: Showed rising use of “gawai” during 2017-2020 while “gadget” fluctuated. The Naive Bayes classifier reached 94% accuracy with a 0.90 macro average.

2024

ACTIVITIES & PROJECTS

Stroke Classification and Severity Estimation from Brain MRI (Undergraduate Thesis)

2025-2026

Built a two-stage PyTorch pipeline using DenseNet121-CBAM for stroke lesion classification (F1 0.8598, AUC 0.9195) and NVAutoNet for 3D lesion segmentation (Dice 0.8347), with lesion volume-based severity estimation.

StrokeAI: Streamlit Application for Stroke MRI Analysis

2026

Built an app that runs preprocessing, classification, 3D segmentation, lesion volume estimation, and severity categorization in one interface, with Clinical MRI and Quick Demo modes.

Sentiment Analysis of PosPay App Reviews with IndoBERT

2025

Scraped 25,000+ Indonesian-language Google Play reviews and cleaned the text with Python.

Labeled sentiment with a pre-trained Indonesian RoBERTa classifier and fine-tuned IndoBERT (PyTorch, Hugging Face Transformers), reaching 89% accuracy and 0.85 macro F1-score on the test set.

Capstone Project: “TemuGigi: Patient Clinical Practice Finder for Dental Students using Images.”

2024

Objective: To develop a mobile app that helps dental students find clinical practice opportunities by using images to detect dental diseases through Convolutional Neural Networks (CNN).

Expected Outcome: The app will improve dental students' practical skills by enabling them to analyze dental disease images, enhancing their diagnostic abilities and overall learning experience.

Bangkit Submission: “Interactive Dashboard Development with Streamlit for Bike Rental Analysis.”

2024

Built an interactive Streamlit dashboard that analyzes rental patterns by season, humidity, and time of day.

COMMITTEE AND VOLUNTEER EXPERIENCES

Fasilkom Fest 2023 - BEM Faculty of Computer Science

Dec 2023 – Feb 2024

Art Division Member, Infographic Contest Head.
Designed the contest theme and coordinated its execution.

Sinau Bareng - Education Office of Surabaya City

2023

Tutor of Sinau Bareng Programme
Mentored literacy and numeracy for primary and junior high school students.

CERTIFICATIONS AND TRAINING

Junior Web Programmer – Badan Nasional Sertifikasi Profesi (BNSP) via LSP UPN

2025

“Veteran” Jawa Timur

Bangkit Academy - led by Google, GoTo, Tokopedia, and Traveloka, Machine Learning Path

Sep 2024 – Dec 2024

TensorFlow Developer Professional Certificate and Machine Learning Specialization

SKILLS

Language	Indonesia (Native Speaker), English (Intermediate)
Programming & Data	Python, R, SQL (MySQL), Pandas, NumPy, Excel, Matplotlib, Seaborn, Streamlit.
Machine Learning	Scikit-learn, TensorFlow, PyTorch, CNN, NLP, Medical Image Segmentation
Statistics & Tools	Descriptive Statistics, Factor Analysis, Validity & Reliability Testing, Time Series Analysis, Git, GitHub, Web Scraping.